

1. A 70-year-old man in cardiac arrest is brought to the emergency department by paramedics. The patient was eating dinner with his wife at a restaurant when he grabbed his chest, slumped over, and passed out. Cardiopulmonary resuscitation was initiated by a bystander and paramedics arrived within 4 minutes. Paramedics say the patient initially had ventricular fibrillation. Electrical defibrillation was attempted once, and he was intubated. An intravenous catheter was inserted, and a single dose of epinephrine was administered 1 minute prior to arrival at the emergency department. On arrival, the patient remains pulseless and apneic. He is placed on a monitor, and a wide-complex, irregular rhythm at a rate of 180/min is noted. In addition to continued ventilation via endotracheal tube, which of the following is the most appropriate next step in management?
- ☐ A) Bedside ECG
  - ☐ B) Electrical defibrillation
  - ☐ C) Intravenous administration of epinephrine
  - ☐ D) Intravenous administration of lidocaine
  - ☐ E) Intravenous administration of vasopressin

2. A 24-year-old man is brought to the emergency department by his friends from a fraternity party. On arrival, he is disheveled and smells of alcohol. He appears sleepy but can be aroused. He is oriented to person but not to place, time, or circumstance. He is unable to provide any further history. His temperature is 36°C (96.8°F), pulse is 95/min, respirations are 12/min, and blood pressure is 100/48 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 92%. Physical examination shows no signs of trauma. Pupils are 5 mm and reactive bilaterally. When he is left alone, he falls asleep with sonorous respirations. Blood alcohol concentration is 280 mg/dL. Which of the following is the most appropriate next step?

- ☐ A) Discharge the patient to the care of a friend
- ☐ B) Monitor the patient's oxygen saturation and end-tidal CO<sub>2</sub> until he is fully oriented and awake
- ☐ C) Observe the patient in the emergency department until his blood alcohol concentration is 0 mg/dL
- ☐ D) Obtain a CT scan of the head
- ☐ E) Place the patient in restraints until he is able to provide further history



3. A 40-year-old man comes to the emergency department because of a 1-day history of low back pain. He reports the pain began after lifting a couch. He has not had fever, weakness, constipation, or urinary retention. He cannot recall any other recent trauma. He has no history of cancer. He does not use illicit drugs. Vital signs are within normal limits. Straight-leg raise test is positive on the right. Sensory function is intact to light touch throughout. Muscle strength is 5/5 in all motor distributions in the lower extremities. Which of the following is the most appropriate next step in management?

- ☐ A) CT scan of the lumbar spine
- ☐ B) Empiric trial of ibuprofen
- ☐ C) Empiric trial of oxycodone
- ☐ D) MRI of the lumbar spine
- ☐ E) X-ray of the lumbar spine



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Lab Values



Calculator



Review



Help



Pause



4. A 20-year-old woman comes to the emergency department 30 minutes after an episode of near syncope when she tried to stand up. She has a 3-day history of progressively worsening shortness of breath and light-headedness. She was diagnosed with acute lymphocytic leukemia 3 weeks ago and is undergoing chemotherapy. On arrival, the patient appears anxious and ill. Her temperature is 37°C (98.6°F), pulse is 124/min, respirations are 28/min, and blood pressure is 70/34 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 98%. Physical examination shows no other abnormalities. An image obtained from bedside ultrasonography of the heart is shown. Which of the following is the most appropriate next step in management?

- ☐ A) Administration of norepinephrine
- ☐ B) Administration of recombinant tissue plasminogen activator
- ☐ C) Broad-spectrum antibiotic therapy
- ☐ D) Intravenous administration of fluids
- ☐ E) Pericardiocentesis



5. A previously healthy 30-year-old woman is brought to the emergency department by her mother because of mental status changes, ringing in her ears, and confusion that began after taking a handful of unknown pills at home 4 hours ago. Her mother says her daughter has been feeling sad and depressed and has had trouble sleeping since her sister died 6 months ago. The patient told her mother she took the pills to "just end it all." The patient appears listless with episodic agitation, is breathing rapidly, and has vomited twice since arrival. Her temperature is 37.8°C (100°F), pulse is 130/min, respirations are 30/min, and blood pressure is 110/60 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 100%. Physical examination shows tremors of the hands. Lungs are clear to auscultation. Laboratory studies show:

|                                    |                          |
|------------------------------------|--------------------------|
| Serum                              |                          |
| Na <sup>+</sup>                    | 144 mEq/L (144 mmol/L)   |
| Cl <sup>-</sup>                    | 101 mEq/L (101 mmol/L)   |
| K <sup>+</sup>                     | 3.0 mEq/L (3 mmol/L)     |
| HCO <sub>3</sub> <sup>-</sup>      | 15 mEq/L (15 mmol/L)     |
| Urea nitrogen                      | 25 mg/dL (8.92 mmol/L)   |
| Glucose                            | 144 mg/dL (7.99 mmol/L)  |
| Creatinine                         | 1.2 mg/dL (106.1 μmol/L) |
| Arterial blood gas analysis shows: |                          |
| pH                                 | 7.34                     |
| Pco <sub>2</sub>                   | 25 mm Hg (3.32 kPa)      |
| Po <sub>2</sub>                    | 88 mm Hg (11.7 kPa)      |

Which of the following is the most appropriate next step in management?

- ☐ A) Gastric lavage
- ☐ B) Intravenous bicarbonate
- ☐ C) Oral activated charcoal
- ☐ D) Oral deferoxamine
- ☐ E) Oral N-acetylcysteine



6. A 38-year-old woman is brought to the emergency department by ambulance because of a 6-hour history of progressive dyspnea with wheezing and dry cough. She has a history of mild persistent asthma treated with inhaled mometasone daily and inhaled albuterol as needed. On arrival, temperature is 37.2°C (99.0°F), pulse is 108/min, respirations are 24/min, and blood pressure is 140/87 mm Hg. Pulse oximetry on 2 L/min of oxygen via nasal cannula is 99%. On physical examination, she appears in mild to moderate respiratory distress but is speaking in long phrases without accessory muscle use. Pulmonary examination discloses diffuse expiratory wheezes bilaterally with prolongation of the expiratory phase. Nebulized short-acting  $\beta$ -adrenergic agonists, nebulized anticholinergics, and parenteral glucocorticoids are initiated, and the patient reports significant improvement in her symptoms after three administrations of nebulized therapy. Which of the following findings, if present, most strongly indicates that this patient can be discharged and effectively treated as an outpatient?
- ☐ A) Audible wheezes are decreased throughout all lung fields
  - ☐ B) Chest x-ray shows normal lung parenchyma and inspiratory volumes
  - ☐ C) Peak expiratory flow rate is 80% of predicted value
  - ☐ D) Pulse rate is less than 100/min





7. A 45-year-old woman is brought to the emergency department by her husband because of a 4-day history of muscle aches, joint pain, and headache and a 1-day history of confusion and severe headache. The husband says that this morning, she could not remember what day it was or what happened to her yesterday. She has inflammatory bowel disease and generalized anxiety disorder. Her medications are sulfasalazine and clonazepam. She has no known drug allergies. She drinks one to two glasses of wine monthly. She does not smoke cigarettes. On arrival, she is oriented to person but not to place or time. She appears uncomfortable. Her eyes are closed; opening her eyes elicits pain. Her temperature is 37.6°C (99.7°F), pulse is 98/min, respirations are 23/min, and blood pressure is 134/78 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 98%. When the neck is passively flexed, she winces in pain and draws her legs up. Cardiopulmonary examination shows no abnormalities. Neurologic examination shows slightly slurred speech. A photograph of the arm is shown. Which of the following is the most appropriate pharmacotherapy?

- ☐ A) Acyclovir
- ☐ B) Amphotericin
- ☐ C) Ampicillin
- ☐ D) Ceftriaxone
- ☐ E) Vancomycin





8. A 68-year-old man is brought to the emergency department by his wife because of a 2-day history of confusion. The patient's wife says that during this time, he has not recognized her and has been talking to people who are not there. This morning she had trouble waking him up, and his speech was completely incoherent. He has alcohol use disorder and drinks one to two pints of vodka daily. His wife reports that his current behavior is nothing like when he is intoxicated. The patient also has hypertension and hyperlipidemia but takes no medications and has not seen a primary care physician in 10 years. Currently, he is obtunded. The patient moans and opens his eyes to sternal rub and withdraws all extremities to painful stimuli. Temperature is 36.8°C (98.2°F), pulse is 72/min, respirations are 18/min, and blood pressure is 144/92 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 97%. Pupils are equal and reactive. Cranial nerves are intact. There is no evidence of trauma. The abdomen is protuberant but soft and nontender. Serum glucose concentration is 118 mg/dL (6.55 mmol/L). CT scan of the head shows no abnormalities. Which of the following studies is most likely to confirm this patient's diagnosis?

- ☐ A) Abdominal ultrasonography
- ☐ B) Diagnostic paracentesis
- ☐ C) Ethanol concentration
- ☐ D) Lumbar puncture
- ☐ E) MRI of the brain
- ☐ F) Serum ammonia concentration



9. A 35-year-old transgender woman comes to the emergency department because of a 2-hour history of chest pain. She reports the pain began suddenly and woke her from sleep. She has a history of hypothyroidism and type 2 diabetes mellitus. Medications are glipizide, levothyroxine, estradiol, and spironolactone. She drinks approximately seven alcoholic beverages weekly and has smoked one-half pack of cigarettes daily for the past 10 years. Her temperature is 37.5°C (99.5°F), pulse is 118/min, respirations are 24/min, and blood pressure is 120/75 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 88%. Physical examination shows moderate edema of the left leg; there is no erythema. Breath sounds are normal. X-ray of the chest shows no abnormalities. ECG shows sinus tachycardia with a right axis deviation. Serum glucose concentration is 120 mg/dL (6.7 mmol/L). Discontinuation of which of the following is most likely to prevent future recurrences of this disease?
- ☐ A) Alcohol
  - ☐ B) Estradiol
  - ☐ C) Glipizide
  - ☐ D) Levothyroxine
  - ☐ E) Spironolactone



10. An 88-year-old man comes to the emergency department because of a 2-month history of severe, intermittent abdominal pain that has increased in severity during the past 3 hours. He associates the pain with eating and has been taking calcium carbonate for indigestion. He has type 2 diabetes mellitus, coronary artery disease, and peripheral vascular disease. He appears to be in severe distress. His temperature is 37.3°C (99.2°F), pulse is 115/min, respirations are 20/min, and blood pressure is 155/94 mm Hg. The abdomen is soft without guarding or rebound; bowel sounds are normal. An increase in which of the following laboratory results is most likely to show a greater severity of this patient's illness?
- ☐ A) Leukocyte count
  - ☐ B) Platelet count
  - ☐ C) Serum bicarbonate concentration
  - ☐ D) Serum glucose concentration
  - ☐ E) Serum lactate concentration



11. A 19-year-old man comes to the emergency department because of a 1-week history of pain and lesions on his penis. He has been sexually active with two new female partners during the past 2 months; they have used condoms inconsistently. The patient has no history of serious illness and takes no medications. Vital signs are within normal limits. Physical examination discloses a group of 1-mm, clear fluid-filled vesicles located midshaft on the dorsum of the penis. The vesicles are tender to touch; no erythema or exudate is noted. Rapid HIV antibody test is negative. Testing for gonorrhea, syphilis, and chlamydia is done and results are pending. In addition to administration of intramuscular ceftriaxone and oral doxycycline, which of the following is the most appropriate next step in management?
- ☐ A) Admit the patient to the hospital for intravenous acyclovir
  - ☐ B) Await results of Tzanck test prior to treatment
  - ☐ C) Discharge the patient with prescription for oral acyclovir
  - ☐ D) Notify the patient's two sexual partners of his diagnosis
  - ☐ E) Notify the public health department of the patient's diagnosis



Previous



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Lab Values



Calculator



Review



Help

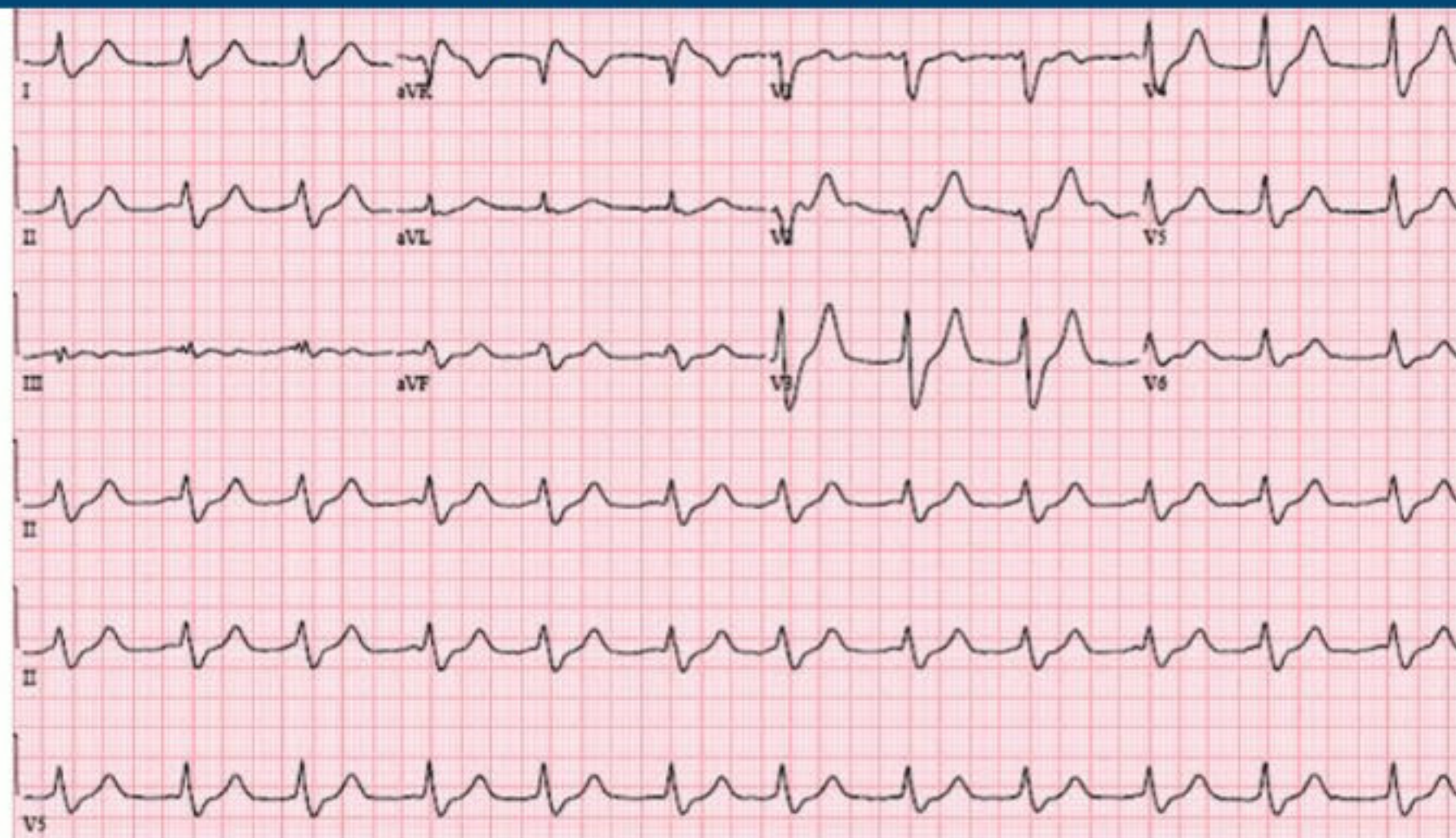


Pause

12. A 2-year-old girl is brought to the emergency department (ED) by her mother because of a 1-week history of fatigue, abdominal pain, and vomiting. She is unable to keep down food or liquids. The patient also has had increased urinary frequency and a 2-kg (4.4-lb) weight loss during this time. The patient's mother brought her to the ED 1 week ago where she was initially diagnosed with "a stomach bug" and did not receive any treatment. Two days later, her symptoms seemed to be getting worse, so her mother brought her to a different ED. At that time, an x-ray of the abdomen was obtained, and the patient was diagnosed with constipation and given an enema. Today, her symptoms have continued to worsen, and the patient's mother reports increased abdominal pain and vomiting. Temperature is 36.8°C (98.2°F), pulse is 130/min, respirations are 45/min, and blood pressure is 65/44 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 100%. The mucous membranes are dry. Lungs are clear to auscultation. The abdomen is soft and there is diffuse tenderness to palpation. Which of the following is the most appropriate next step in diagnosis?

- ☐ A) Chest x-ray
- ☐ B) Complete blood count
- ☐ C) Fingertstick blood glucose concentration
- ☐ D) Ultrasonography of the abdomen
- ☐ E) Urinalysis





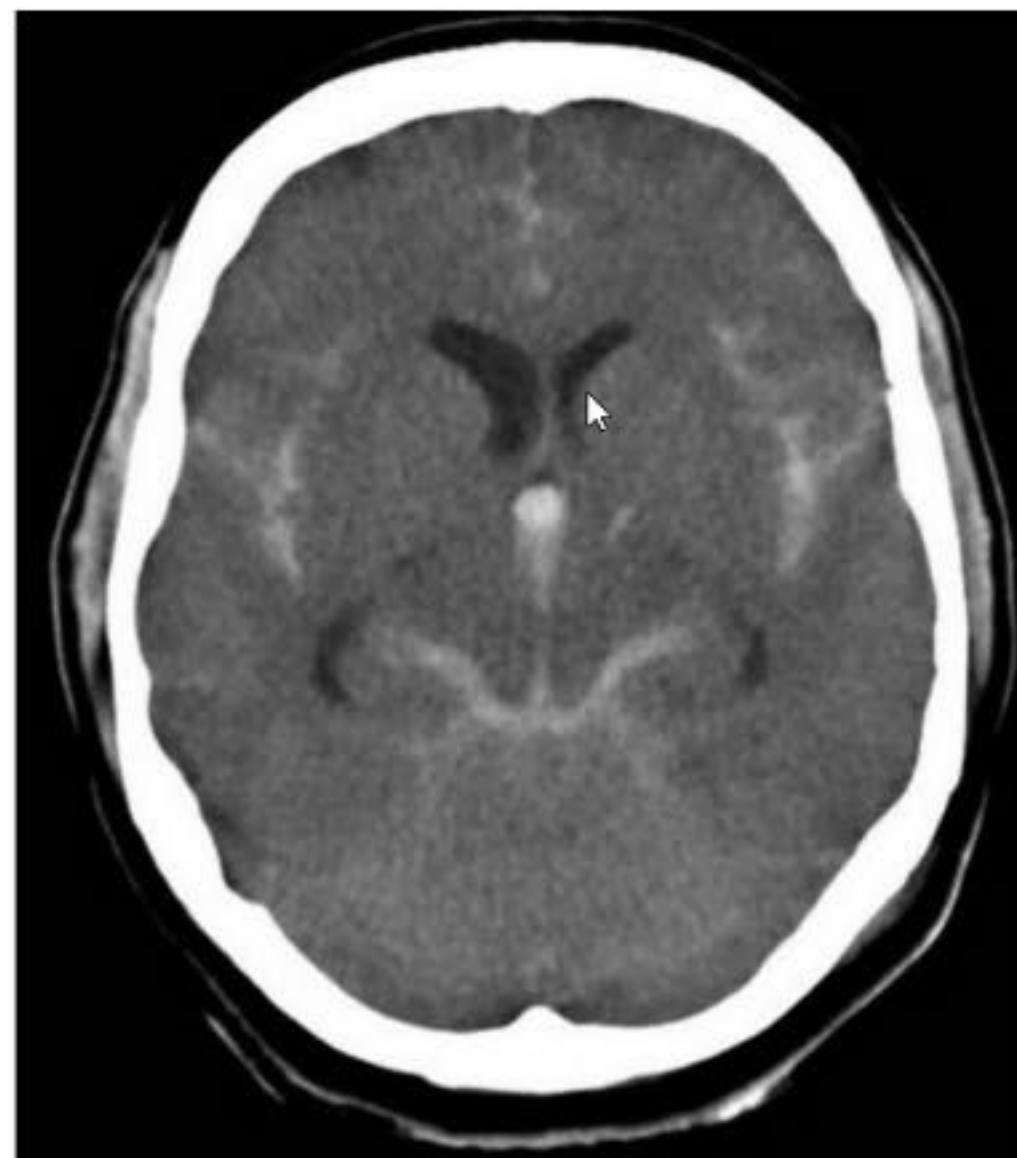
13. A 68-year-old man with end-stage renal disease is brought to the emergency department at 8 PM on a Saturday because he has missed his last three hemodialysis appointments because of a major snowstorm and closure of his usual dialysis center. He also has hypertension treated with amlodipine and lisinopril. On arrival, he has mild shortness of breath but is otherwise asymptomatic. Vital signs are within normal limits. Pulse oximetry on room air shows an oxygen saturation of 95%. Auscultation of the lungs discloses faint bilateral basilar crackles. An ECG is shown. Which of the following is the most appropriate next step in management?

- ☐ A) Administer calcium gluconate, insulin, and dextrose; schedule hemodialysis on Monday
- ☐ B) Administer calcium gluconate, insulin, and dextrose; schedule hemodialysis within 2 hours
- ☐ C) Administer sodium polystyrene sulfonate (Kayexalate); schedule hemodialysis on Monday
- ☐ D) Administer sodium polystyrene sulfonate (Kayexalate); schedule hemodialysis within 2 hours



14. An independently living widowed 90-year-old woman is brought to the emergency department (ED) by paramedics 15 minutes after collapsing in a store. En route to the ED, the patient is nasally intubated by paramedics because of altered mental status. On arrival, temperature is 36.0°C (96.8°F), pulse is 42/min, respirations are ventilated at 16/min, and blood pressure is 210/106 mm Hg. Pulse oximetry on  $\text{FiO}_2$  of 0.5 shows an oxygen saturation of 100%. CT scan of the head shows a massive intracranial hemorrhage with areas of subfalcine herniation. Multiple family members arrive at the ED and report that the patient and her family were aware of a previously diagnosed large cerebral artery aneurysm. The patient's only child further states the patient had completed a do-not-resuscitate/do-not-intubate order; however, the do-not-resuscitate order cannot be located. Which of the following is the most appropriate next step in management?
- ☐ A) Consult the hospital ethics committee
  - ☐ B) Consult the hospital legal department
  - ☐ C) Conversion to orotracheal intubation
  - ☐ D) Extubation
  - ☐ E) Operative treatment





15. A 60-year-old man is brought to the emergency department 2 hours after the sudden onset of severe headache, nausea, and vomiting. He has chronic renal failure secondary to type 1 diabetes mellitus. He has received hemodialysis for 2 years. His last dialysis was 2 days ago. His only medication is insulin. On arrival, the patient is somnolent and moaning in pain. His temperature is 37°C (98.6°F), pulse is 80/min, respirations are 22/min, and blood pressure is 140/90 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 98%. Physical examination shows spontaneous eye opening. The pupils are 3 mm and reactive to light. Laboratory studies show:

|                  |                                                    |
|------------------|----------------------------------------------------|
| Hemoglobin       | 9.5 g/dL (95 g/L)                                  |
| Leukocyte count  | 12,600/mm <sup>3</sup> (12.6 × 10 <sup>9</sup> /L) |
| Platelet count   | 90/mm <sup>3</sup> (0.09 × 10 <sup>9</sup> /L)     |
| Prothrombin time | 14 sec (INR=1.1)                                   |





15. A 60-year-old man is brought to the emergency department 2 hours after the sudden onset of severe headache, nausea, and vomiting. He has chronic renal failure secondary to type 1 diabetes mellitus. He has received hemodialysis for 2 years. His last dialysis was 2 days ago. His only medication is insulin. On arrival, the patient is somnolent and moaning in pain. His temperature is 37°C (98.6°F), pulse is 80/min, respirations are 22/min, and blood pressure is 140/90 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 98%. Physical examination shows spontaneous eye opening. The pupils are 3 mm and reactive to light. Laboratory studies show:

|                               |                                                    |
|-------------------------------|----------------------------------------------------|
| Hemoglobin                    | 9.5 g/dL (95 g/L)                                  |
| Leukocyte count               | 12,600/mm <sup>3</sup> (12.6 × 10 <sup>9</sup> /L) |
| Platelet count                | 90/mm <sup>3</sup> (0.09 × 10 <sup>9</sup> /L)     |
| Prothrombin time              | 14 sec (INR=1.1)                                   |
| Partial thromboplastin time   | 30 sec                                             |
| Serum                         |                                                    |
| Na <sup>+</sup>               | 145 mEq/L (145 mmol/L)                             |
| K <sup>+</sup>                | 5.2 mEq/L (5.2 mmol/L)                             |
| HCO <sub>3</sub> <sup>-</sup> | 21 mEq/L (21 mmol/L)                               |
| Cl <sup>-</sup>               | 103 mEq/L (103 mmol/L)                             |
| Urea nitrogen                 | 88 mg/dL (31.4 mmol/L)                             |
| Creatinine                    | 4.1 mg/dL (362.4 μmol/L)                           |
| Glucose                       | 135 mg/dL (7.49 mmol/L)                            |

A CT scan of the head is shown. Administration of which of the following is the most appropriate next step in management?

- ☐ A) Fresh frozen plasma
- ☐ B) Packed red blood cells
- ☐ C) Platelets
- ☐ D) Vitamin K
- ☐ E) Whole blood





16. A 6-year-old boy is brought to the emergency department by his parents 30 minutes after being bitten on the hand by the family cat. The patient has a history of exercise-induced asthma, for which he takes inhaled albuterol as needed. Both the patient's and the cat's immunizations are up-to-date. Temperature is 37.0°C (98.6°F), pulse is 90/min, respirations are 18/min, and blood pressure is 99/66 mm Hg. Physical examination discloses a puncture wound to the palmar aspect of the left hand. There is minimal swelling and tenderness to palpation over the affected area. Active and passive range of motion is full and muscle strength is 5/5 in all digits. The remainder of the physical examination discloses no other abnormalities. Without treatment, this patient is most likely to develop an infection from which of the following pathogens?

- ☐ A) *Bacteroides fragilis*
- ☐ B) *Bartonella henselae*
- ☐ C) *Eikenella corrodens*
- ☐ D) *Pasteurella multocida*
- ☐ E) *Staphylococcus aureus*
- ☐ F) *Streptococcus pyogenes*



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Lab Values



Calculator



Review



Help



Pause



17. A 42-year-old man comes to the emergency department because of a 3-day history of left leg pain and swelling. He has not had a recent injury and reports no other symptoms. The patient's medical history includes hyperlipidemia, a laparoscopic cholecystectomy performed 6 months ago, and deep venous thrombosis treated 2 years ago. His only medication is atorvastatin. He has smoked one pack of cigarettes daily for the past 20 years and drinks alcoholic beverages socially. BMI is 31 kg/m<sup>2</sup>. Temperature is 37.2°C (99.0°F), pulse is 88/min, respirations are 14/min, and blood pressure is 136/76 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 97%. Physical examination discloses tenderness to palpation and edema of the left lower extremity but no other abnormalities. Compression ultrasonography shows a noncompressible left popliteal vein. Which of the following factors in this patient's history most increased his risk for developing this condition?

- ☐ A) BMI of 31 kg/m<sup>2</sup>
- ☐ B) Deep venous thrombosis
- ☐ C) Hyperlipidemia
- ☐ D) Surgical history
- ☐ E) Tobacco use



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Lab Values



Calculator



Review



Help



Pause





Mark

18. A 35-year-old man comes to the emergency department 30 minutes after he fell from a ladder and sustained a laceration of the right forearm. Vital signs are within normal limits. Physical examination shows a deep laceration over the lateral aspect of the right mid forearm, and fat globules are visibly evident in the fluid coming from the wound. Which of the following is the most likely explanation for these findings?

- ☐ A) Compartment syndrome
- ☐ B) Fat embolus
- ☐ C) Muscle laceration
- ☐ D) Nerve injury
- ☐ E) Open fracture



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Review



Help



Pause



19. A 63-year-old man comes to the emergency department 30 minutes after an episode of syncope. He suffered no trauma during the episode. The patient says he was jogging on a treadmill at a gym when he felt dizzy and light-headed and had a brief fainting episode. He joined the gym 3 weeks ago to lose weight and "get in shape." He reports having had a similar episode last week when he felt faint and dizzy, but says he did not pass out. He also has had mild chest pain and shortness of breath with exertion, but attributes it to "being out of shape". He has a history of hypertension but adheres poorly to medications and reports not seeing a doctor for the past 10 years. He smoked one pack of cigarettes daily for the past 30 years but quit 2 years ago. The patient is awake, alert, and in no distress. Temperature is 37.4°C (99.3°F), pulse is 92/min, respirations are 16/min, and blood pressure is 178/86 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 96%. Mild crackles are heard at both lung bases. The abdomen is soft and nontender to palpation. No edema of the lower extremities is noted. Which of the following is the most likely underlying cause of this patient's symptoms?

- ☐ A) Aortic stenosis
- ☐ B) Mitral regurgitation
- ☐ C) Pulmonic stenosis
- ☐ D) Tricuspid regurgitation
- ☐ E) Ventricular septal defect



20. An unconscious 37-year-old man is brought to the emergency department 45 minutes after being submerged in an ice-covered freshwater lake for 10 minutes. At the scene, his temperature was 32°C (89.6°F); cardiopulmonary resuscitation (CPR) was started. En route, he was intubated and mechanically ventilated; CPR was continued, and he was covered in warm blankets. On arrival, he is unresponsive to painful stimuli, and CPR is continued. His temperature is 33.7°C (92.7°F). During compressions, there is no spontaneous pulse. His palpable systolic blood pressure is 60 mm Hg. As CPR continues, he remains intubated and is ventilated at a rate of 12/min. Orotracheal suction yields a large amount of water. Forty-five minutes after arrival, the patient has a generalized tonic-clonic seizure. Which of the following is the most likely cause of this seizure?

- ☐ A) Hypercalcemia
- ☐ B) Hyperkalemia
- ☐ C) Hyponatremia
- ☐ D) Hypocalcemia
- ☐ E) Hypokalemia
- ☐ F) Hyponatremia









21. A 23-year-old man comes to the emergency department 30 minutes after his vehicle fell on his right leg. The patient was working underneath his vehicle when the jack slipped and the vehicle fell on top of his right leg; he was stuck for 10 minutes before bystanders were able to remove the vehicle. The patient reports experiencing severe pain in his right leg, which he rates as 10 on a 10-point scale. He has no history of serious illness and takes no medications. On arrival, temperature is 37.0°C (98.6°F), pulse is 118/min, respirations are 22/min, and blood pressure is 143/85 mm Hg. Physical examination discloses swelling and bruising of the right lower extremity. Active movement of the ankle is limited by severe pain. Passive movement of the right great toe elicits pain. Capillary refill time is 6 seconds and pulses are diminished in the right lower extremity. X-rays of the right ankle are shown. Which of the following is the most appropriate next step in management?

- ☐ A) Administer intravenous heparin
- ☐ B) Consult a surgeon
- ☐ C) Measure ankle brachial indices
- ☐ D) Obtain duplex ultrasonography of the right lower extremity
- ☐ E) Position the patient in bed so that his injured leg is hanging over the edge of the bed



22. A 42-year-old woman comes to the emergency department because of a 2-hour history of vomiting. During this time, the patient has experienced 10 episodes of emesis and noted several streaks of bright red blood during the most recent episode. She has not experienced any abdominal or chest pain but continues to feel nauseated. She has not experienced similar episodes of bleeding in the past. The patient says that she drank eight alcoholic beverages the previous night. She has a history of hypothyroidism, for which she takes levothyroxine. Temperature is 37.0°C (98.6°F), pulse is 85/min, respirations are 18/min, and blood pressure is 120/73 mm Hg. Physical examination discloses no abnormalities. Intravenous ondansetron and 0.9% saline are administered with improvement of the patient's nausea. Which of the following is the most likely diagnosis?

- ☐ A) Arteriovenous malformation
- ☐ B) Boerhaave syndrome
- ☐ C) Esophageal varices
- ☐ D) Mallory-Weiss tear
- ☐ E) Peptic ulcer disease



23. A 75-year-old man is transferred from a nursing care facility to the emergency department because of a 24-hour history of altered mental status and fever. He has dementia, Alzheimer type, and sustained a stroke 3 years ago. He has an indwelling catheter because of urinary incontinence. According to the nursing care facility staff, the patient is typically alert and oriented to person and place; during the past 24 hours, he has not been oriented to person, place, or time. On arrival, the patient is awake but remains disoriented. Temperature is 38.5°C (101.3°F), pulse is 78/min, respirations are 22/min, and blood pressure is 110/75 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 100%. There is no apparent skin breakdown. Lungs are clear to auscultation; no crackles are heard. The abdomen is soft and nontender. An indwelling urinary catheter with a drainage bag is in place. Urinalysis is shown:

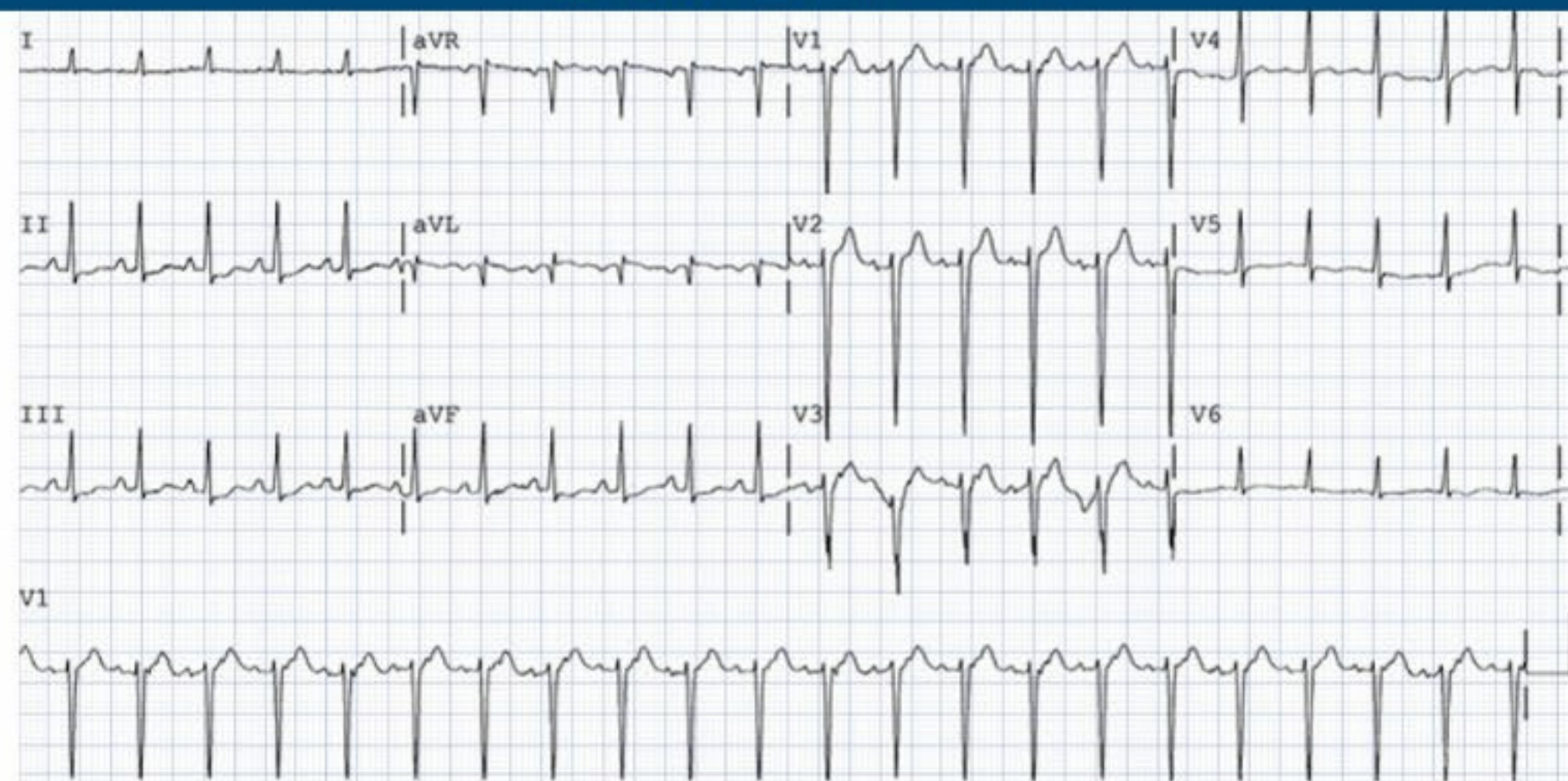
|            |          |
|------------|----------|
| Appearance | cloudy   |
| WBC        | >100/hpf |
| Nitrites   | positive |
| Bacteria   | positive |

Chest x-ray shows no abnormalities. Which of the following is the most appropriate next step in management?

- ☐ A) Irrigation of the catheter with water twice daily
- ☐ B) Prophylactic antibiotic therapy
- ☐ C) Removal of the catheter
- ☐ D) Replacement of the catheter every 2 days
- ☐ E) Replacement of the leg drainage bag daily

24. A 25-year-old primigravid woman at 15 weeks' gestation comes to the emergency department because of a 24-hour history of urinary frequency. She has not had vomiting, fever, flank pain, vaginal bleeding, or abnormal vaginal discharge. She has no history of serious illness, and her only medication is a prenatal vitamin. Temperature is 36.9°C (98.4°F), pulse is 75/min, respirations are 18/min, and blood pressure is 100/65 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 100%. Physical examination shows no abnormalities. Which of the following clean-catch urinalysis results is most likely to require pharmacotherapy in this patient?
- ☐ A) Bacteria
  - ☐ B) Crystals
  - ☐ C) Ketones
  - ☐ D) Protein
  - ☐ E) Squamous epithelial cells





25. A 45-year-old man with a history of cocaine use is brought to the emergency department because of severe pain in the middle of his chest that began suddenly 20 minutes ago after using cocaine. He is otherwise healthy and takes no medications. His temperature is 37°C (98.6°F), pulse is 130/min, respirations are 20/min, and blood pressure is 220/110 mm Hg. Physical examination shows no other abnormalities. ECG is shown. Which of the following is the most appropriate pharmacotherapy?

- ☐ A) Heparin
- ☐ B) Lorazepam
- ☐ C) Morphine
- ☐ D) Nitroglycerin
- ☐ E) Tissue plasminogen activator





26. A 32-year-old woman is brought to the emergency department by her friend because of a 30-minute history of facial swelling and shortness of breath that began after she was stung by a bee. The patient has a history of asthma and eczema. Her only medication is albuterol as needed. Temperature is 36.9°C (98.4°F), pulse is 118/min, respirations are 28/min, and blood pressure is 88/40 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 91%. Physical examination shows periorbital and lip edema as well as a diffuse, urticarial rash. Examination of the posterior pharynx is limited by tongue edema. Faint, high-pitched sounds are heard with inspiration. In addition to epinephrine therapy, which of the following is the most appropriate next step in management?

- ☐ A) Bilevel positive airway pressure
- ☐ B) Diphenhydramine therapy
- ☐ C) Endotracheal intubation
- ☐ D) Methylprednisolone therapy
- ☐ E) Supplemental oxygen therapy



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Calculator



Review



Help



Pause





27. A 17-year-old girl is brought to the emergency department because of the gradual onset of headache during the past 2 weeks. She says that the headache is worse in the mornings, accompanied by vomiting and sensitivity to light, and does not respond to treatment with ibuprofen and acetaminophen. She has not had fever, rash, neck pain, or changes in vision. She has never had headaches like this previously. She has a history of primary hydrocephalus for which a ventriculoperitoneal shunt was placed 5 years ago. She has routine follow-up appointments with her neurosurgeon and has never needed revision or replacement of the shunt. She has no other history of operative procedures. There is no family history of migraines. She takes no medications. She does not smoke cigarettes, drink alcohol, or use illicit drugs. Immunizations are up-to-date. On arrival, the patient is in moderate distress. Her temperature is 37.7°C (99.9°F), pulse is 50/min, respirations are 24/min, and blood pressure is 170/65 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 99%. A shunt reservoir is palpable on the scalp behind the left ear. No erythema is noted. The pupils are 4 mm, equal, and reactive to light. The fundi are not well visualized. Cardiopulmonary examination shows no abnormalities. Strength and sensation are full, and the cranial nerves are intact. Neurologic examination shows difficulty with tandem gait and finger-nose testing bilaterally. Which of the following is the most likely mechanism of this patient's condition?

- ☐ A) Increased vascular permeability
- ☐ B) Infection
- ☐ C) Mechanical obstruction
- ☐ D) Toxic exposure
- ☐ E) Vasospasm





28. A 9-year-old boy is brought to the emergency department by his mother because of a 2-day history of inability to put full weight on his left leg. The patient says that he was playing basketball prior to the onset of the pain but does not recall any specific trauma. He has no history of serious illness and receives no medications. He is 137 cm (4 ft 6 in; 75th percentile) tall and weighs 47 kg (104 lb; 95th percentile). His temperature is 36.7°C (98.1°F), pulse is 90/min, respirations are 16/min, and blood pressure is 118/68 mm Hg. On physical examination, the patient walks with a limp favoring the left side. Internal and external rotation of the left hip elicits pain. No abnormalities of the overlying skin are noted. There is no tenderness of the spine. Distal neurovascular examination shows no abnormalities. An x-ray of the pelvis is shown. Which of the following is the most likely cause of these findings?

- ☐ A) Legg-Calvé-Perthes disease
- ☐ B) Osgood-Schlatter disease
- ☐ C) Septic arthritis
- ☐ D) Slipped capital femoral epiphysis
- ☐ E) Synovitis



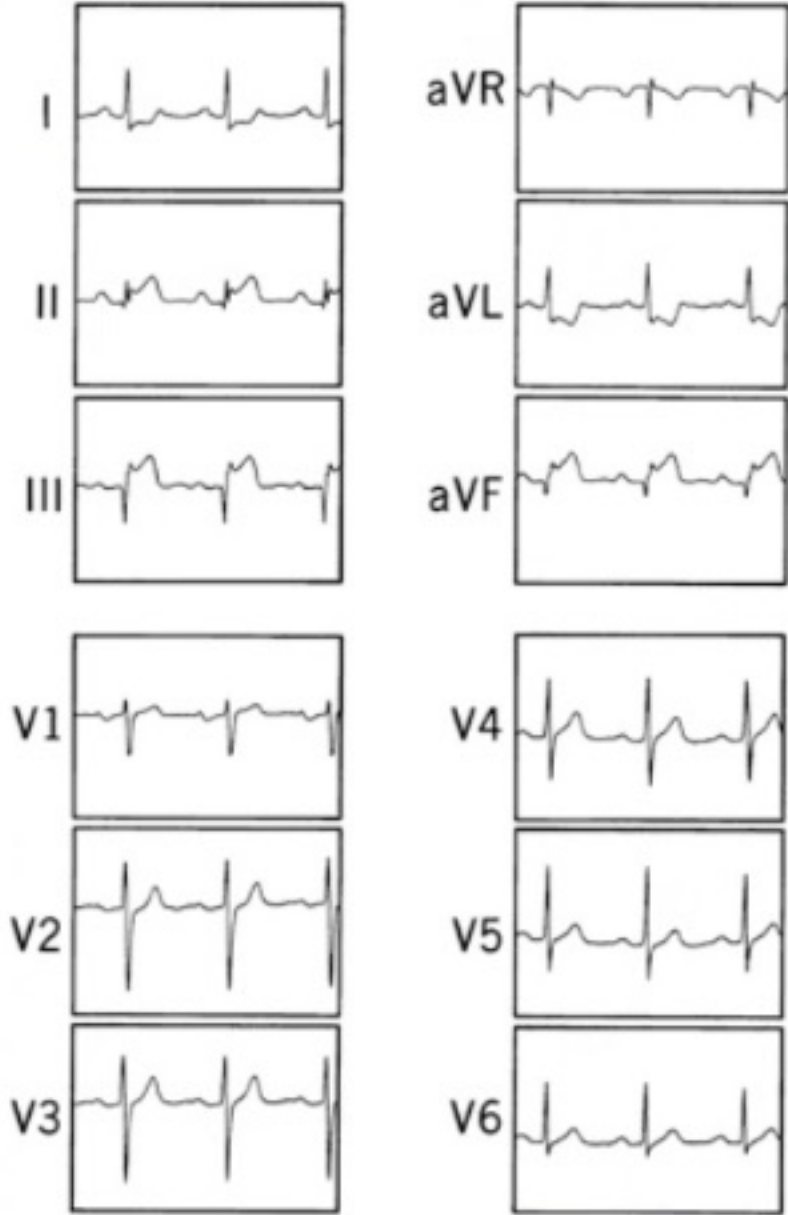
☐ Mark

29. A 75-year-old man comes to the emergency department because of a 1-week history of headache and nausea. He has been housebound during the past 2 weeks because of a hip fracture and noted the headache gradually worsening during this time. The patient says, "I've never had a headache like this before." He has not had any fever, visual symptoms, neck stiffness, vomiting, abdominal pain, or rash. His hip fracture was sustained as the result of a mechanical fall at home and was repaired without complication 1 day following the injury. After a 2-day hospital stay, he was discharged. The patient also has a history of hypertension. His current medications are enoxaparin, metoprolol, hydrocodone-acetaminophen, and docusate. Temperature is 36.9°C (98.4°F), pulse is 69/min, respirations are 16/min, and blood pressure is 165/75 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 100%. Physical examination discloses no nuchal rigidity or rash. Cardiopulmonary and neurologic examinations disclose no abnormalities. The surgical incision is clean, dry, and intact without any erythema or exudate. ECG discloses a normal sinus rhythm at 70/min with no ST- or T-wave abnormalities. CT scan of the head discloses no abnormalities. Complete blood count, complete metabolic panel, and serum troponin concentration are within the reference ranges. Which of the following is the most appropriate diagnostic study at this time?
- ☐ A) MRI of the brain
  - ☐ B) Prothrombin time
  - ☐ C) Serum acetaminophen concentration
  - ☐ D) Serum carboxyhemoglobin



30. A 60-year-old woman comes to the emergency department because of a 2-hour history of chest pain and shortness of breath. The symptoms began suddenly while she was at work; the patient says the pain is located at the center of her chest and does not radiate. She has hypertension and type 2 diabetes mellitus. Three months ago, she traveled to Europe. One month ago, she was diagnosed with bronchitis, which improved after treatment with azithromycin. Current medications are enalapril and metformin. She has smoked five to six cigarettes daily for the past 30 years. On arrival, the patient is pale and diaphoretic; she appears to be in mild distress. Temperature is 36.1°C (97.0°F), pulse is 50/min, respirations are 20/min, and blood pressure is 100/60 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 98%. Physical examination shows no abnormalities. ECG is shown. Which of the following is the most likely cause of this patient's symptoms?

- ☐ A) Blockage of conduction through the atrioventricular node
- ☐ B) Dissection of the ascending aorta
- ☐ C) Occlusion of the left pulmonary artery
- ☐ D) Plaque rupture in the right coronary artery
- ☐ E) Viral inflammation of the myocardium





31. A 65-year-old man comes to the emergency department because of a 2-day history of shortness of breath. He has chronic obstructive pulmonary disease treated with prednisone and inhaled albuterol. He had smoked two packs of cigarettes daily for 40 years but quit 1 year ago. His temperature is 37.5°C (99.5°F), pulse is 100/min, respirations are 20/min, and blood pressure is 130/90 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 94%. Auscultation of the lungs discloses bilateral wheezes. No crackles or rhonchi are audible. The patient is treated with nebulized albuterol-  
ipratropium. Which of the following is the most appropriate recommendation to prevent exacerbation of this patient's condition?

- ☐ A) Adoption of an exercise regimen
- ☐ B) Influenza virus vaccination annually
- ☐ C) Pneumococcal polysaccharide vaccine, 23-valent, vaccination annually
- ☐ D) Prophylactic azithromycin therapy
- ☐ E) Prophylactic home oxygen therapy at night



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Lab Values



Calculator



Review



Help



Pause





32. A 22-year-old man is brought to the emergency department by ambulance 30 minutes after sustaining multiple gunshot wounds to the chest. The patient reports severe chest pain and difficulty breathing. He has no history of serious illness, takes no medications or other substances, and has no known allergies. He does not smoke cigarettes or drink alcoholic beverages. Temperature is 37.0°C (98.6°F), pulse is 132/min, respirations are 30/min, and blood pressure is 82/48 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 86%. Physical examination shows two gunshot wounds to the right chest. Breath sounds are decreased on the right. An emergent needle decompression is performed on the right chest, which results in adequate air return, an increase in oxygen saturation to 96%, and improved breathing. Pulse rate and blood pressure are unchanged; a chest tube is placed and immediately yields 1600 mL of blood. Which of the following is the most appropriate next step in management?

- ☐ A) Endotracheal intubation
- ☐ B) Pericardiocentesis
- ☐ C) Placement of a central venous catheter
- ☐ D) Placement of a second chest tube
- ☐ E) Surgical exploration



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Calculator



Review



Help



Pause





33. A 75-year-old man is brought to the emergency department because of worsening abdominal pain that began suddenly 1 day ago. He has had five episodes of vomiting and 10 episodes of diarrhea during this time. He has not had fever. He has a history of hypercholesterolemia, type 2 diabetes mellitus, and hypertension. He sustained a myocardial infarction 10 years ago. His medications are atorvastatin, metformin, and lisinopril. He has not recently taken antibiotics and has no known sick contacts. His temperature is 37°C (98.6°F), pulse is 140/min and irregular, respirations are 24/min, and blood pressure is 165/90 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 98%. The lungs are clear to auscultation. The abdomen is soft; there is mild, diffuse tenderness to palpation without rebound tenderness or guarding. No masses or organomegaly is noted. Rectal examination shows brown stool; test of the stool for occult blood is positive. There is no pedal edema. Laboratory studies show:

|                               |                                                    |
|-------------------------------|----------------------------------------------------|
| Hematocrit                    | 43%                                                |
| Leukocyte count               | 14,300/mm <sup>3</sup> (14.3 × 10 <sup>9</sup> /L) |
| Platelet count                | 340,000/mm <sup>3</sup> (340 × 10 <sup>9</sup> /L) |
| Serum                         |                                                    |
| Na <sup>+</sup>               | 140 mEq/L (140 mmol/L)                             |
| K <sup>+</sup>                | 3.9 mEq/L (3.9 mmol/L)                             |
| Cl <sup>-</sup>               | 100 mEq/L (100 mmol/L)                             |
| HCO <sub>3</sub> <sup>-</sup> | 13 mEq/L (13 mmol/L)                               |
| Urea nitrogen                 | 20 mg/dL (7.1 mmol/L)                              |
| Creatinine                    | 1.4 mg/dL (123.8 μmol/L)                           |
| Lipase                        | 40 U/L [N=14–280]                                  |
| Lactic acid                   | 5.2 mEq/L (5.2 mmol/L) [N<2 mEq/L//5.2 mmol/L]     |

An ECG shows atrial fibrillation at a rate of 140/min. Which of the following is the most appropriate next step in management?

- ☐ A) Blood cultures
- ☐ B) CT scan of the abdomen
- ☐ C) Electrical cardioversion
- ☐ D) Intravenous antibiotic therapy
- ☐ E) Measurement of serum troponin concentration



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Help



Pause



34. A 19-year-old woman comes to the emergency department because of a 2-day history of mild lower abdominal pain, urinary urgency, and a burning sensation during urination. The patient also reports a chronic history of minimal clear vaginal discharge. She has not had fever, nausea, or vomiting. She has no history of serious illness and her only medication is an oral contraceptive. She has a penicillin allergy. The patient has been sexually active with one male partner during the past year. They use condoms consistently. Vital signs are within normal limits. Physical examination shows a soft, nondistended abdomen. There is mild suprapubic tenderness to palpation with no rebound or guarding. Urine pregnancy test is negative. Urinalysis is positive for nitrites and leukocyte esterase. Which of the following is the most appropriate pharmacotherapy?

- ☐ A) Azithromycin
- ☐ B) Cephalexin
- ☐ C) Clindamycin
- ☐ D) Metronidazole
- ☐ E) Trimethoprim-sulfamethoxazole



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Help



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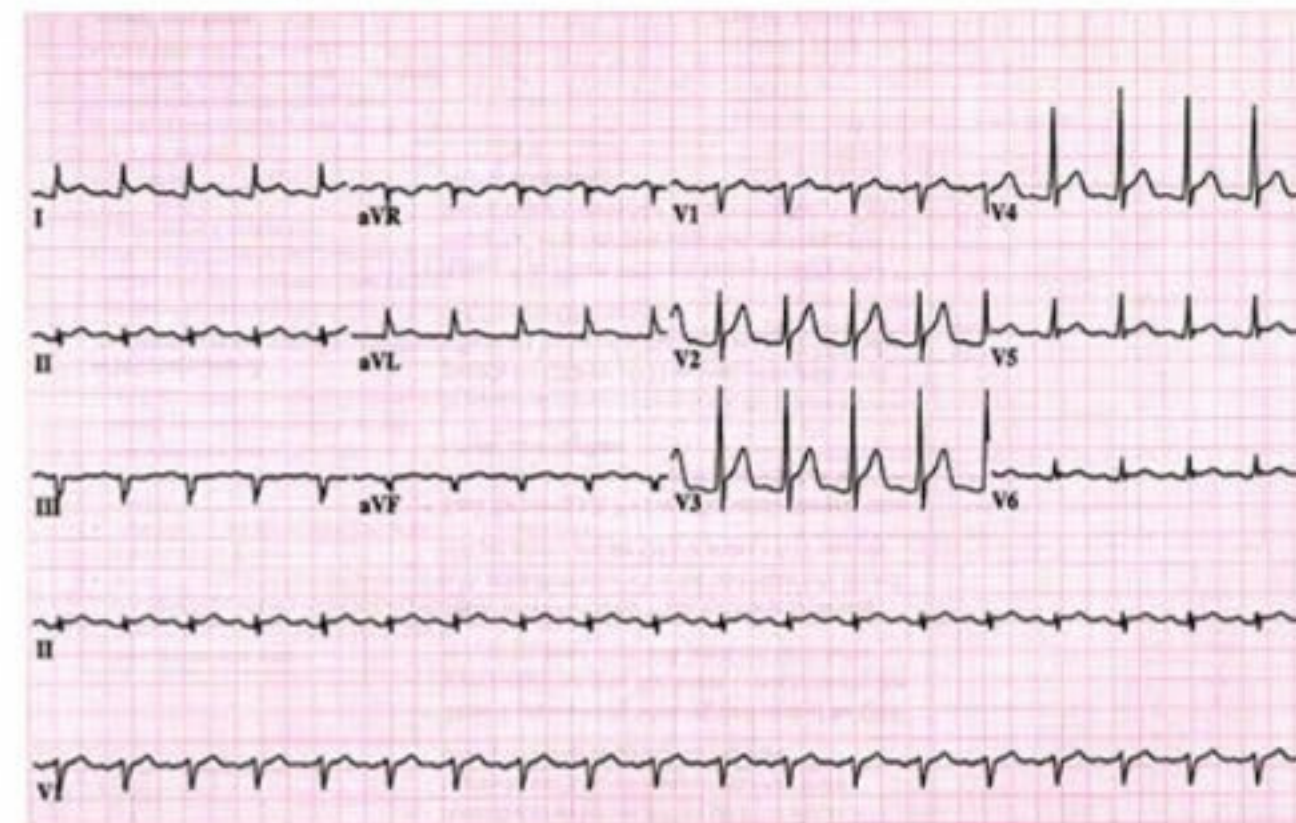
35. A 15-year-old girl is brought to the emergency department by her father because of a 3-day history of progressive weakness and a 1-day history of moderate abdominal cramps and nausea. She says that she was too weak to get out of bed and go to school today. She has not had fever or diarrhea. She has no history of serious illness and takes no medications. She has no known drug allergies. BMI is 19 kg/m<sup>2</sup>. Her temperature is 37.2°C (99°F), pulse is 114/min and regular, respirations are 24/min, and blood pressure is 114/72 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 98%. The mucous membranes are dry. The lungs are clear to auscultation. The abdomen is soft; there is mild epigastric tenderness to palpation. The remainder of the examination shows no abnormalities. Serum studies show:

|                               |                        |
|-------------------------------|------------------------|
| Na <sup>+</sup>               | 131 mEq/L (131 mmol/L) |
| K <sup>+</sup>                | 3.1 mEq/L (3.1 mmol/L) |
| Cl <sup>-</sup>               | 95 mEq/L (95 mmol/L)   |
| HCO <sub>3</sub> <sup>-</sup> | 7 mEq/L (7 mmol/L)     |
| Urea nitrogen                 | 23 mg/dL (8.2 mmol/L)  |
| Glucose                       | 685 mg/dL (38 mmol/L)  |
| Creatinine                    | 1.3 mg/dL (115 μmol/L) |

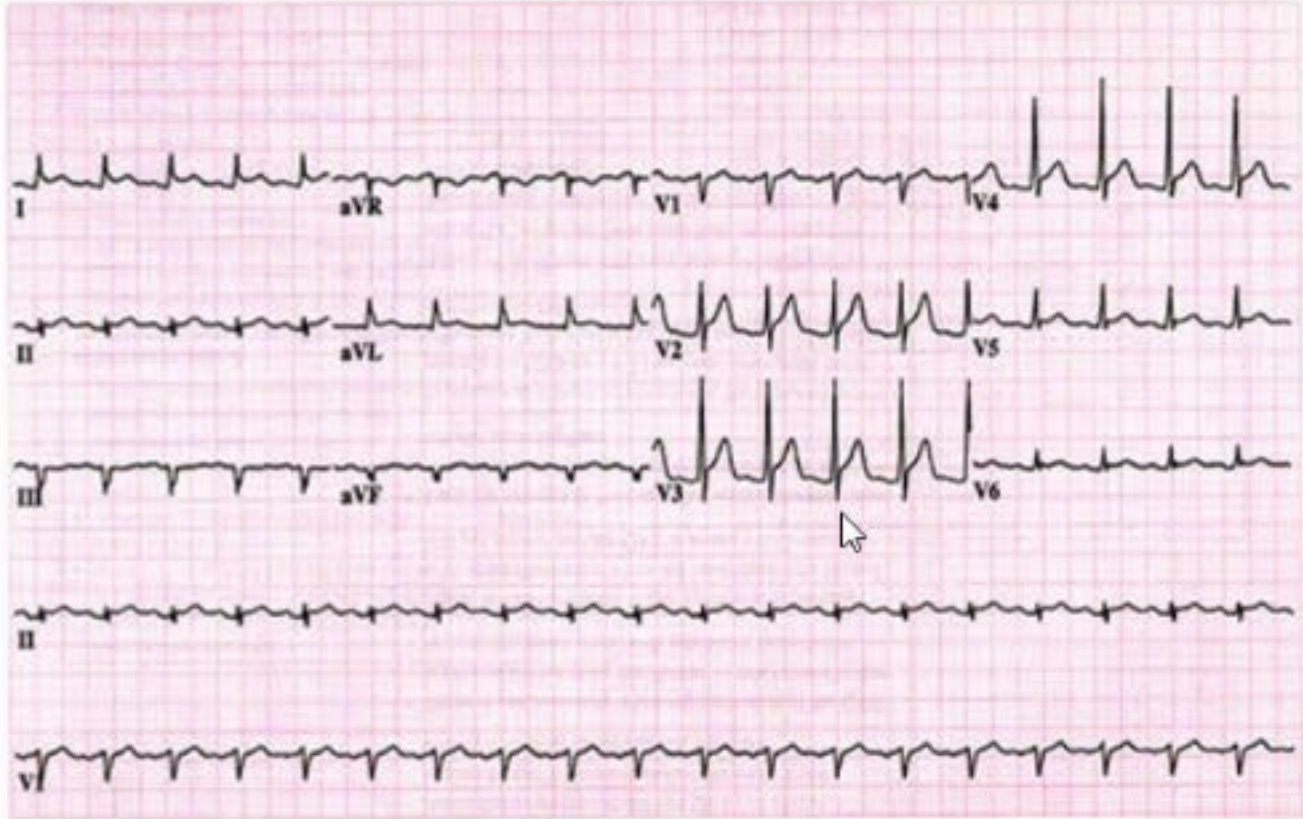
Which of the following is the most appropriate next step in management?

- ☐ A) Administration of bicarbonate
- ☐ B) Administration of insulin
- ☐ C) Administration of 0.9% saline
- ☐ D) Administration of sodium citrate
- ☐ E) Hemodialysis









36. A 28-year-old man comes to the emergency department because of a 10-hour history of sharp, substernal chest pain that awoke him from sleep. The pain improves when he leans forward and worsens with inspiration and when he is supine. During the past 5 days, he has had dry cough, runny nose, sore throat, diffuse muscle aches, fatigue, malaise, and temperatures to 37.9°C (100.2°F). He has not had shortness of breath. He has asthma treated with inhaled albuterol as needed. He has never been hospitalized for asthma and takes no daily medications. He underwent appendectomy at age 7 years. He does not smoke cigarettes, drink alcoholic beverages, or use illicit drugs. He has no known drug allergies. His temperature is 37.7°C (99.8°F), pulse is 124/min, respirations are 16/min, and blood pressure is 110/74 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 97%. Physical examination shows no abnormalities. A chest x-ray and ECG are shown. Which of the following is the most appropriate pharmacotherapy?
- ☐ A) Aspirin
  - ☐ B) Azithromycin
  - ☐ C) Ibuprofen
  - ☐ D) Oseltamivir
  - ☐ E) Verapamil



37. A 29-year-old woman comes to the emergency department because of a 3-day history of intermittent palpitations and diarrhea, and a 1-week history of "jitters," nausea, fatigue, and generalized weakness. The patient emigrated from El Salvador to the United States 5 years ago. Using an interpreter, she reports a history of "thyroid problems" and says that she used to take an unknown medication to control her condition before she immigrated to the United States. She has not had access to health care since immigrating and ran out of her medication 2 weeks ago. She is unsure if she has had a fever but notes that she has been sweating with minimal exertion. Temperature is 38.1°C (100.6°F), pulse is 132/min, respirations are 22/min, and blood pressure is 128/70 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 99%. Physical examination shows no abnormalities. Serum TSH concentration is 0.09 µU/mL. Which of the following is the most appropriate pharmacotherapy?

- ☐ A) Iodine
- ☐ B) Levothyroxine
- ☐ C) Methimazole
- ☐ D) Propranolol
- ☐ E) Propylthiouracil



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Lab Values



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Review



Help



Pause



38. An 82-year-old woman is transferred to the emergency department from a nursing care facility because of a 1-day history of a fever. She has atrial fibrillation, congestive heart failure, anemia, stroke, and dementia, Alzheimer type. Medications are metoprolol, lisinopril, clopidogrel, furosemide, daily aspirin, and an iron supplement. The patient arouses to voice but mumbles incoherently. Temperature is 38.4°C (101.1°F), pulse is 115/min and irregular, respirations are 18/min, and blood pressure is 92/64 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 96%. Physical examination shows no signs of trauma to the head; the neck is supple. Lungs are clear to auscultation. Cardiac examination discloses no murmurs. The abdomen is soft with mild suprapubic tenderness. The percutaneous endoscopic gastrostomy site is clear. There is a stage 1 sacral decubitus ulcer. Portable chest x-ray shows bilateral basilar atelectasis. ECG shows atrial fibrillation. Laboratory studies show:

|                               |                                                    |
|-------------------------------|----------------------------------------------------|
| Hemoglobin                    | 9.6 g/dL (96 g/L)                                  |
| Hematocrit                    | 28%                                                |
| Leukocyte count               | 18,100/mm <sup>3</sup> (18.1 × 10 <sup>9</sup> /L) |
| Platelet count                | 375,000/mm <sup>3</sup> (375 × 10 <sup>9</sup> /L) |
| Serum                         |                                                    |
| Na <sup>+</sup>               | 131 mEq/L (131 mmol/L)                             |
| Cl <sup>-</sup>               | 100 mEq/L (100 mmol/L)                             |
| K <sup>+</sup>                | 4.2 mEq/L (4.2 mmol/L)                             |
| HCO <sub>3</sub> <sup>-</sup> | 14 mEq/L (14 mmol/L)                               |
| Urea nitrogen                 | 18 mg/dL (6.43 mmol/L)                             |
| Glucose                       | 81 mg/dL (4.5 mmol/L)                              |
| Creatinine                    | 1.2 mg/dL (106.1 μmol/L)                           |
| Lactic acid                   | 4.3 mEq/L (4.3 mmol/L)                             |
| Urinalysis                    |                                                    |
| Specific gravity              | 1.031                                              |
| Blood                         | negative                                           |
| Glucose                       | negative                                           |
| Protein                       | 2+                                                 |
| WBC                           | 3+                                                 |
| Nitrites                      | negative                                           |
| Leukocyte esterase            | positive                                           |
| RBC                           | 0–5/hpf                                            |
| WBC                           | 50–100/hpf                                         |

Which of the following is the most likely cause of this patient's current condition?



|                               |                                                    |
|-------------------------------|----------------------------------------------------|
| Hemoglobin                    | 9.6 g/dL (96 g/L)                                  |
| Hematocrit                    | 28%                                                |
| Leukocyte count               | 18,100/mm <sup>3</sup> (18.1 × 10 <sup>9</sup> /L) |
| Platelet count                | 375,000/mm <sup>3</sup> (375 × 10 <sup>9</sup> /L) |
| Serum                         |                                                    |
| Na <sup>+</sup>               | 131 mEq/L (131 mmol/L)                             |
| Cl <sup>-</sup>               | 100 mEq/L (100 mmol/L)                             |
| K <sup>+</sup>                | 4.2 mEq/L (4.2 mmol/L)                             |
| HCO <sub>3</sub> <sup>-</sup> | 14 mEq/L (14 mmol/L)                               |
| Urea nitrogen                 | 18 mg/dL (6.43 mmol/L)                             |
| Glucose                       | 81 mg/dL (4.5 mmol/L)                              |
| Creatinine                    | 1.2 mg/dL (106.1 μmol/L)                           |
| Lactic acid                   | 4.3 mEq/L (4.3 mmol/L)                             |
| Urinalysis                    |                                                    |
| Specific gravity              | 1.031                                              |
| Blood                         | negative                                           |
| Glucose                       | negative                                           |
| Protein                       | 2+                                                 |
| WBC                           | 3+                                                 |
| Nitrites                      | negative                                           |
| Leukocyte esterase            | positive                                           |
| RBC                           | 0–5/hpf                                            |
| WBC                           | 50–100/hpf                                         |

Which of the following is the most likely cause of this patient's current condition?

- ☐ A) Altered osmotic gradient caused by a decreased serum sodium concentration
- ☐ B) Decreased oxygen carrying capacity of the blood
- ☐ C) Impaired cardiac output caused by decreased ventricular filling
- ☐ D) Vasodilation of circulatory beds



39. A 45-year-old woman comes to the emergency department (ED) to request a refill for hydromorphone. The patient has a 10-year history of chronic pain in her right knee and says that her current pain is not new or different. She rates her current pain as a 10 on a 10-point scale. She reports no recent trauma or unusual activity. She has no other medical history. In addition to hydromorphone, her current medications are trazodone, lorazepam, amitriptyline, and hydroxyzine. She states she took her last dose of hydromorphone prior to her arrival today, but she has enough of her other medications. She has an appointment with her pain specialist tomorrow. Medical records disclose that the patient has had five pain-related visits to the ED during the past 2 months. Temperature is 37.0°C (98.6°F), pulse is 88/min, respirations are 18/min, and blood pressure is 135/80 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 96%. The pupils measure 2 mm bilaterally and are equally reactive to light. Cardiopulmonary examination discloses no abnormalities. No edema, erythema, or effusion of the right knee is noted, but passive and active range of motion is limited by pain. Distal pulses in the lower extremities are normal. Which of the following is the most appropriate next step in management?
- ☐ A) Administer hydromorphone
  - ☐ B) Consult orthopaedics
  - ☐ C) Contact the patient's pain specialist
  - ☐ D) Discharge the patient with a 3-day course of hydromorphone
  - ☐ E) Obtain x-ray of the right knee



40. A 74-year-old woman is transferred from a skilled nursing care facility to the emergency department because of a 6-hour history of altered mental status. The nursing staff reports that the patient is typically alert and talkative, but her speech is now difficult to understand and she has been falling to her right side when walking; she ambulates with a walker. She has a history of dementia, Alzheimer type; hypertension; and diverticulitis. She had a cerebral infarction 4 years ago with no residual symptoms. Current medications include memantine, haloperidol, and lisinopril. She is allergic to aspirin. On arrival, she is alert and oriented to person but not to place or time. Her temperature is 37.1°C (98.8°F), pulse is 84/min, respirations are 14/min, and blood pressure is 190/100 mm Hg. Examination shows drooping of the right side of the face and slurred speech. Muscle strength is 4/5 in the right upper and lower extremities. CT scan of the head shows hypodensity in the territory of the left middle cerebral artery, which is unchanged from a CT scan of the head 2 years ago; there are no acute findings. Which of the following therapeutic interventions is most likely to have prevented this patient's current condition?

- ☐ A) Adding clopidogrel
- ☐ B) Adding low-dose trimethoprim-sulfamethoxazole
- ☐ C) Adding apixaban
- ☐ D) Increasing the dose of lisinopril
- ☐ E) Switching from haloperidol to ziprasidone



Previous



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Lab Values



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Review



Help



Pause





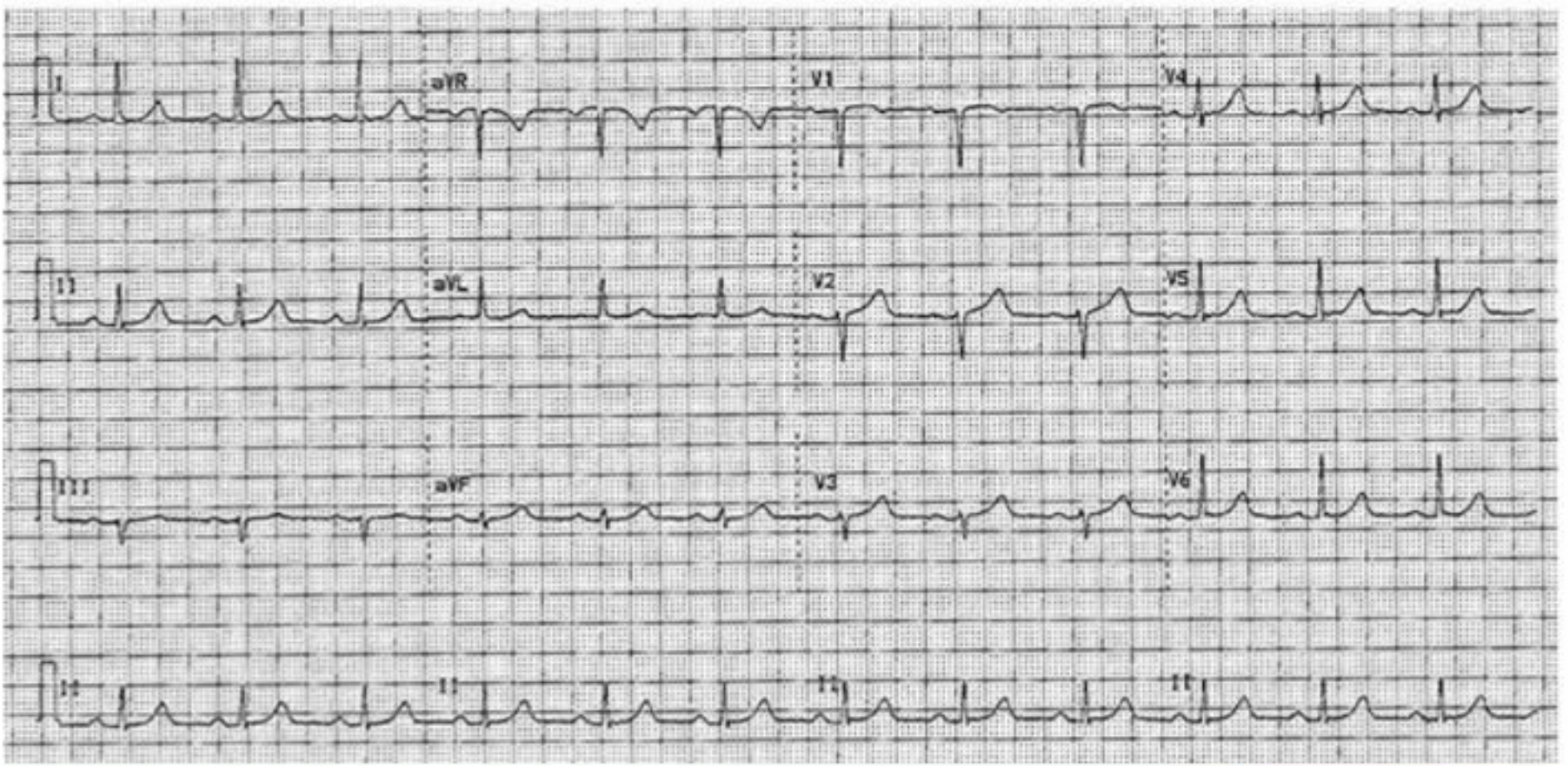
41. A 32-year-old man comes to the emergency department because of a 3-day history of progressively severe pain, swelling, and redness in his left calf; he believes he "must have been bitten by a spider." He has not had fever. He has no history of serious illness and takes no medications. The patient does not use intravenous drugs. Vital signs are within normal limits. Physical examination shows an area of erythema and induration over the left calf which measures 10 cm in diameter; there is mild tenderness to palpation over the area. No subcutaneous air is palpable, no bullae or ecchymoses are noted, and there is no lymphangitic streaking or proximal lymphadenopathy. Ultrasonography of the affected area is shown. Which of the following is the most appropriate next step in management?
- ☐ A) Administration of antibiotics
  - ☐ B) CT scan of the lower extremity
  - ☐ C) Incision and drainage
  - ☐ D) MRI of the lower extremity
  - ☐ E) Observation only



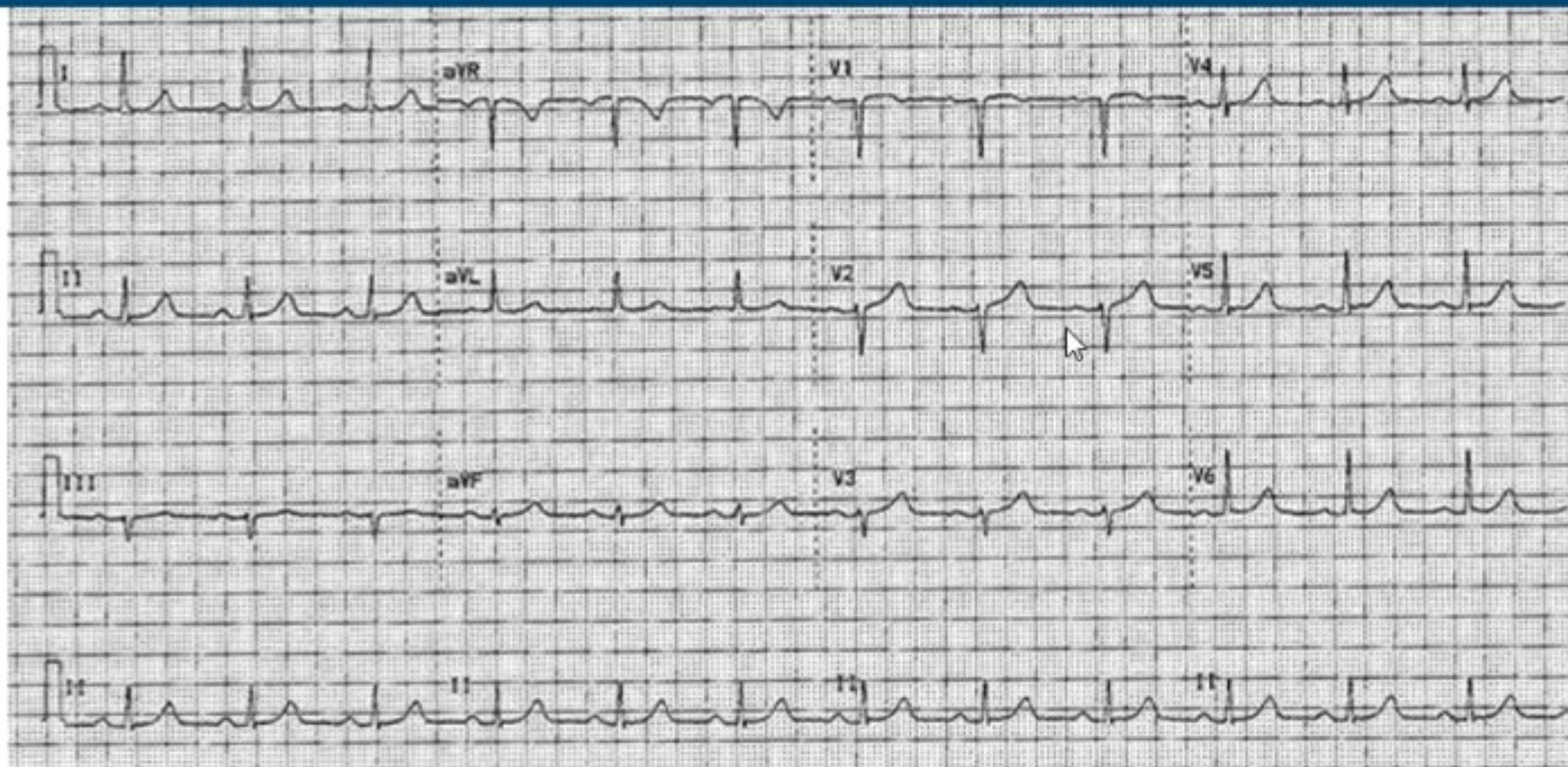
42. A 17-year-old boy is brought to the emergency department by his parents 20 minutes after he attempted suicide by ingesting liquid drain cleaner. On arrival, he has chest and abdominal pain. He is awake and alert but agitated. He insists that he be left alone to die. Examination shows a pale gray exudate on the lips and oral mucosa. The lungs are clear to auscultation. Abdominal examination shows epigastric tenderness. Which of the following is the most appropriate next step in diagnosis?

- ☐ A) Measurement of arterial blood pH
- ☐ B) Measurement of gastric pH
- ☐ C) Upper gastrointestinal series
- ☐ D) CT scan of the chest
- ☐ E) Esophagogastroduodenoscopy









43. A 68-year-old man is brought to the emergency department by his wife because of a 1-hour history of sharp, moderate chest pain that radiates to his back and generalized weakness. He has not had shortness of breath or sweating. He has hypertension treated with amlodipine. He has no known allergies. He has smoked one pack of cigarettes daily for 40 years. He does not use illicit drugs. The patient is in mild distress. BMI is 23 kg/m<sup>2</sup>. His temperature is 37.1°C (98.8°F), pulse is 86/min, respirations are 18/min, and blood pressure is 164/94 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 98%. The abdomen is soft and nontender. Femoral pulses are normal. The remainder of the examination, including cardiopulmonary examination, shows no abnormalities. A chest x-ray and ECG are shown. Administration of which of the following is the most appropriate next step?

- ☐ A) Heparin
- ☐ B) Labetalol
- ☐ C) Nitroglycerin
- ☐ D) Oxygen
- ☐ E) Tissue plasminogen activator





Mark

44. A 77-year-old man is brought to the emergency department after being rescued from a house fire. Firefighters found him unconscious in his bed; the bedroom was full of smoke. The firefighters and paramedics state that the patient probably fell asleep while smoking in bed. On arrival, he does not respond to verbal stimuli and has cough productive of carbonaceous sputum. Temperature is 37.8°C (100.0°F), pulse is 120/min, respirations are 30/min and labored, and blood pressure is 98/78 mm Hg. Pulse oximetry on 50% oxygen by face mask shows an oxygen saturation of 93%. Examination shows second-degree burns over the face and upper extremities. There is marked facial edema, and the eyebrows are singed. On pulmonary examination, air movement is poor. Stridor and wheezes are heard over the lung bases bilaterally. In addition to intravenous administration of fluids, which of the following is the most appropriate next step in management?

- ☐ A) Bronchoalveolar lavage
- ☐ B) Continuous positive airway pressure
- ☐ C) Endotracheal intubation
- ☐ D) Inhaled nitric oxide therapy
- ☐ E) Tracheostomy



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45. A 51-year-old woman comes to the emergency department because of a 1-week history of fever, fatigue, and a nonproductive cough that causes sharp, right-sided, pleuritic chest pain. Since yesterday, her cough has become productive of clear-to-yellow sputum. She has no history of serious illness and takes no medications. She has smoked one-half pack of cigarettes daily for the past 25 years. She has not recently traveled, taken antibiotics, or been hospitalized, and she has had no known sick contacts. Temperature is 38.0°C (100.4°F), pulse is 105/min, respirations are 16/min, and blood pressure is 115/85 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 100%. She appears well, speaks in full sentences, and is not in respiratory distress. Pulmonary examination discloses fine crackles in the right mid-lung field; the remainder of the examination shows no abnormalities. A chest x-ray is shown. Which of the following is the most appropriate next step in management?

☐ A) Admit and administer ceftriaxone





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- ☐ A) Admit and administer ceftriaxone
  - ☐ B) Admit and administer levofloxacin and prednisone
  - ☐ C) Discharge and prescribe amoxicillin and azithromycin
  - ☐ D) Discharge and prescribe trimethoprim-sulfamethoxazole



46. A 65-year-old man comes to the emergency department because of a 2-hour history of vomiting dark blood. He also has had light-headedness and sweating during this time. He has a medical history of alcoholic cirrhosis. Medications are lactulose, metoprolol, and hydrochlorothiazide. On arrival, the patient is pale and diaphoretic; he appears to be in moderate distress. Temperature is 36.7°C (98.1°F), pulse is 120/min, respirations are 24/min, and blood pressure is 80/50 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 94%. Physical examination shows dried blood on the lips and tongue. The remainder of the examination shows no abnormalities. Following intravenous administration of 2 L of 0.9% saline, the patient's vital signs are unchanged. Which of the following is the most appropriate next step in management?

- ☐ A) Endotracheal intubation
- ☐ B) Intravenous administration of 1 L of lactated Ringer solution
- ☐ C) Intravenous administration of phenylephrine
- ☐ D) Placement of a nasogastric tube
- ☐ E) Transfusion of packed red blood cells





47. A 32-year-old primigravid woman at 32 weeks' gestation comes to the emergency department because of a 1-week history of achy upper abdominal pain and swelling of her legs. The symptoms have become increasingly severe during the past 24 hours. She has not had shortness of breath, vomiting, diarrhea, or leg pain. Her prenatal course has been otherwise uncomplicated. Her temperature is 36.7°C (98°F), pulse is 100/min, respirations are 20/min, and blood pressure is 170/90 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 97%. Abdominal examination shows mild tenderness to palpation of the right upper quadrant. A photograph of the left lower extremity is shown. Which of the following is most likely to confirm the diagnosis?

- ☐ A) ECG
- ☐ B) Liver function testing
- ☐ C) Measurement of serum B-type natriuretic peptide concentration
- ☐ D) Serum D-dimer assay
- ☐ E) Urine toxicology screening





48. A 28-year-old woman comes to the emergency department because of a 3-day history of steadily worsening nonradiating lower abdominal pain. She rates the pain as an 8 on a 10-point scale. She also has had a 1-week history of white vaginal discharge. She has not had dysuria or urinary urgency or frequency. The patient has no history of serious illness and takes no medications. She has no known allergies. Menses occur at 26- to 30-day intervals; she does not recall the date of her last menstrual period. She is sexually active with multiple male partners, and they use condoms inconsistently. Her temperature is 37.4°C (99.4°F), pulse is 88/min, respirations are 14/min, and blood pressure is 104/72 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 99%. The abdomen is soft, with tenderness in the bilateral lower quadrants without rebound or guarding. Pelvic examination discloses thick white discharge in the vaginal vault and cervical motion tenderness. No adnexal masses or tenderness are noted. Urine pregnancy test is negative. Cervical swabs are positive for chlamydia and negative for gonorrhea. Which of the following is the most appropriate treatment?

- ☐ A) Metronidazole and ceftriaxone
- ☐ B) Ciprofloxacin and doxycycline
- ☐ C) Clindamycin and azithromycin
- ☐ D) Doxycycline and ceftriaxone
- ☐ E) Fluconazole and metronidazole



49. A 55-year-old woman comes to the emergency department 30 minutes after she sustained a burn to her hand by accidentally grabbing a hot pan while cooking. She rinsed her hand under cool water prior to arrival. She has a history of hypertension and type 2 diabetes mellitus for which she takes metoprolol and metformin. Her last tetanus vaccination was 3 years ago. Temperature is 36.5°C (97.7°F), pulse is 79/min, respirations are 20/min, and blood pressure is 169/75 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 100%. Physical examination discloses a full-thickness burn to the entire right palm and palmar aspects of the index finger, middle finger, ring finger, and small finger. Blistering is present; some blisters are intact while others have ruptured. No erythema or exudate is noted. Capillary refill time is less than 2 seconds in the tips of all digits. Sensation distal to the wound is intact to light touch. Ketamine is administered and the blisters are debrided. Which of the following is the most appropriate next step in management?
- ☐ A) Intravenous bolus of 0.9% saline
  - ☐ B) Oral cephalexin therapy
  - ☐ C) Topical bacitracin and bandaging
  - ☐ D) X-ray of the hand



50. A 54-year-old woman comes to the emergency department (ED) because of a 4-hour history of right upper quadrant pain. Medical history includes type 2 diabetes mellitus, hypertension, and coronary artery disease. She is diagnosed with acute cholecystitis and is scheduled to be transferred to the operating room. The physician who initially evaluated the patient signs her out to another physician, reporting she is only "waiting for surgery, nothing to do." For the next 8 hours the patient is held because of emergent cases arriving in the ED. During a follow-up assessment, the patient is found to be minimally responsive. Numerous providers rush to the bedside, and it is determined that the patient's finger-stick blood glucose concentration is 36 mg/dL (2 mmol/L). Intravenous dextrose is administered and the patient's condition improves. Which of the following is most likely to decrease the recurrence of this error?

- ☐ A) Having the oncoming physician evaluate each patient
- ☐ B) Prominently displaying signs above the beds of patients with diabetes mellitus indicating their condition
- ☐ C) Scheduling blood glucose checks for any patient boarding in the ED
- ☐ D) Using a structured sign-out tool



Previous



Next



Lab Values



Calculator



Review



Help



Pause